A BROWN REPRESENTABILITY CRITERION FOR FUNCTORS COMMUTING WITH LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3872-3916, following the statement at 3851-3870.

Editorial note. Footnote at 3853-3854 points to remark-big-categories for the set-theoretic size convention.

Allowed standard prerequisites (editorial).

- Limits exist in C. Hypothesis of the lemma (big category with limits); the limit defining x is taken at proof line 3878. Existence of x = lim x_i and of the equalizers used later.
- F commutes with limits. Hypothesis, applied at proof lines 3881-3885 and 3904-3905. F(x) = lim F(x_i) gives the universal element f; commutation with equalizers gives f' and f".
- Yoneda-style transformation construction. Section section-opposite of this chapter, invoked at proof lines 3888-3892. The transformation xi: Mor(x, -) -> F(-) induced by the universal element.

Source reference. Original remark 0015: sources/stacks-categories/pinned-original-source.tex, source lines 294–294. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 001L: sources/stacks-categories/pinned-original-source.tex, source lines 655–655. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a big[1] category which has limits. Let $F : \mathcal{C} \to Sets$ be a functor. Assume that*

(1) *F commutes with limits,*
(2) *there exist a family $\{x_i\}_{i \in I}$ of objects of $\mathcal{C}$ and for each $i \in I$ an element $f_i \in F(x_i)$ such that for $y \in \mathrm{Ob}(\mathcal{C})$ and $g \in F(y)$ there exist an i and a morphism $\varphi : x_i \to y$ with $F(\varphi)(f_i) = g$.*

Then F is representable, i.e., there exists an object x of $\mathcal{C}$ such that

$$F(y) = \mathrm{Mor}_{\mathcal{C}}(x, y)$$

functorially in y.

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[1]See Remark 0015.

Proof. Let $\mathcal{I}$ be the category whose objects are the pairs (x_i, f_i) and whose morphisms $(x_i, f_i) \to (x_{i'}, f_{i'})$ are maps $\varphi : x_i \to x_{i'}$ in $\mathcal{C}$ such that $F(\varphi)(f_i) = f_{i'}$. Set

$$x = \lim_{(x_i, f_i) \in \mathcal{I}} x_i$$

(this will not be the x we are looking for, see below). The limit exists by assumption. As F commutes with limits we have

$$F(x) = \lim_{(x_i, f_i) \in \mathcal{I}} F(x_i).$$

Hence there is a universal element $f \in F(x)$ which maps to $f_i \in F(x_i)$ under F applied to the projection map $x \to x_i$. Using f we obtain a transformation of functors

$$\xi : \mathrm{Mor}_{\mathcal{C}}(x, -) \longrightarrow F(-)$$

see Section 001L. Let y be an arbitrary object of $\mathcal{C}$ and let $g \in F(y)$. Choose $x_i \to y$ such that f_i maps to g which is possible by assumption. Then F applied to the maps

$$x \longrightarrow x_i \longrightarrow y$$

(the first being the projection map of the limit defining x) sends f to g. Hence the transformation ξ is surjective.

In order to find the object representing F we let $e : x' \to x$ be the equalizer of all self maps $\varphi : x \to x$ with $F(\varphi)(f) = f$. Since F commutes with limits, it commutes with equalizers, and we see there exists an $f' \in F(x')$ mapping to f in $F(x)$. Since ξ is surjective and since f' maps to f we see that also $\xi' : \mathrm{Mor}_{\mathcal{C}}(x', -) \to F(-)$ is surjective. Finally, suppose that $a, b : x' \to y$ are two maps such that $F(a)(f') = F(b)(f')$. We have to show $a = b$. Consider the equalizer $e' : x'' \to x'$. Again we find $f'' \in F(x'')$ mapping to f'. Choose a map $\psi : x \to x''$ such that $F(\psi)(f) = f''$. Then we see that $e \circ e' \circ \psi : x \to x$ is a morphism with $F(e \circ e' \circ \psi)(f) = f$. Hence $e \circ e' \circ \psi \circ e = e$. Since e is a monomorphism, this implies that e' is an epimorphism, thus $a = b$ as desired. $\square$